1. Battery in and E-stop bank B (LM5069-1, two FETs). Bank A, in series ahead of it, is block 7

Pack: 8 AWG pigtail pair to an XT90 (mates the pack XT90S), 1 BAT, 2 GND

2. Control rail and E-stop latch (no logic IC)

12.1 V, 11 mA. On the battery side, so regen through the body diodes keeps it up with the BMS open. Runs the latch and the clamp. 0.5 W at 54.6 V. Fed through F1 with its own 1 uF, clamped by a 15 V TVS, reverse diode OUT to IN: block 10.

2a. Pi heartbeat watchdog: no heartbeat, no latch

3. Braking clamp: 56.6–56.8 V on, 55.8–55.9 V off (typ), 57.5 V worst case, into 2 x HS50 20R in parallel

4. Bus capacitors

5. 12 V converter: own LM5146 buck board (marengo_buck12) on the -X rail, pigtail J24 (block 10); DVE0125V2NBC is the drop-in fallback

5a. Brick soft start (ahead of the E-stop switch)

6. Limb ports: 1.5 mOhm shunt, own breaker (block 6a), meter, soldered pigtail pair. Pigtail fuses inline.

6a. Limb port breakers: TPS48111-Q1 + one FET per port, latch-off, enabled by RUN

7. E-stop bank A: second series bank, own driver, ahead of bank B

8. Spare 12 V outputs

8a. SPARE_ON eFuse

9. E-stop pressed: button NO contact to the Pi

10. Rev B2 protection parts (refs appended so earlier refs keep their numbers)

10b. Absorber_close option (M3): 0R fitted, the rest DNP

10c. Rev B4 review parts

10d. Rev B5 parts

Rev B: a part goes in only if leaving it out would hurt. Parts list: hardware/electrical/pdb/REV_B_PAR

marengo

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Title: marengo power board rev B

Size: A0	Date: 2026-11-06	**Rev: B-wip**
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